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A population approach to cortical GABAergic interneuron function

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https://doi.org/10.1016/j.neuron.2026.01.028

Inhibitory interneuron diversity is a central feature of cortical circuits. The IN-CODE consortium seeks to combine large-scale recordings of interneuron types with machine-learning tools to identify the role of their physiological features, connectivity motifs, and cooperativity in cognitive functions.

Introduction

GABAergic inhibitory interneurons are strikingly diverse components of cortical circuits that control virtually all aspects of pyramidal cell function. While inhibition was conceptualized as a "sculpting process" for the encoding of information in the brain nearly 50 years ago, the question of how this sculpting occurs and how it should be addressed experimentally and analytically have been hotly debated ever since. In the first major wave of research, many scientists focused on the identification of individual interneuron types and performed a detailed analysis of their physiological, morphological, and molecular features (Figure 1A). This seminal "single-neuron approach" generated detailed taxonomies of GABAergic cells and has recently received a considerable boost through the rise of high-throughput methods such as connectomics, transcriptomics, proteomics, and metabolomics, which allow neuronal characterization at various scales. Although the single-neuron approach is largely pursued *in vitro*, an *in vivo* thrust emerged while dissecting interneuron function at the microcircuit level. This has been fueled by the generation of transgenic mouse lines and the development of viral vectors for gene delivery to defined interneuron classes (e.g., parvalbumin [PV]-, somatostatin [SOM]-, and vasoactive intestinal peptide [VIP]-expressing cells) to progress large-scale activity recordings of interneuron and pyramidal cell populations using one- and two-photon calcium imaging or extracellular single-unit recordings (e.g., silicon probe arrays, Neuropixels). The inception of optogenetic and chemogenetic perturbations of the activity in selected interneuron types further provided information on their computational roles in neuronal circuits. This battery of methods ushered in the era of the "microcircuit approach" in the early 2010s. Since then, this type of research has revolutionized our understanding of how the different types of interneurons are functionally integrated in local information processing circuits and, ultimately, how they shape the representation of environmental features and events on the level of individual cells, neuronal associations, and animal behavior (Figure 1B).

Current challenges in interneuron research

Both the neuron and the microcircuit approaches are indispensable for systems neuroscience and continue to be of key

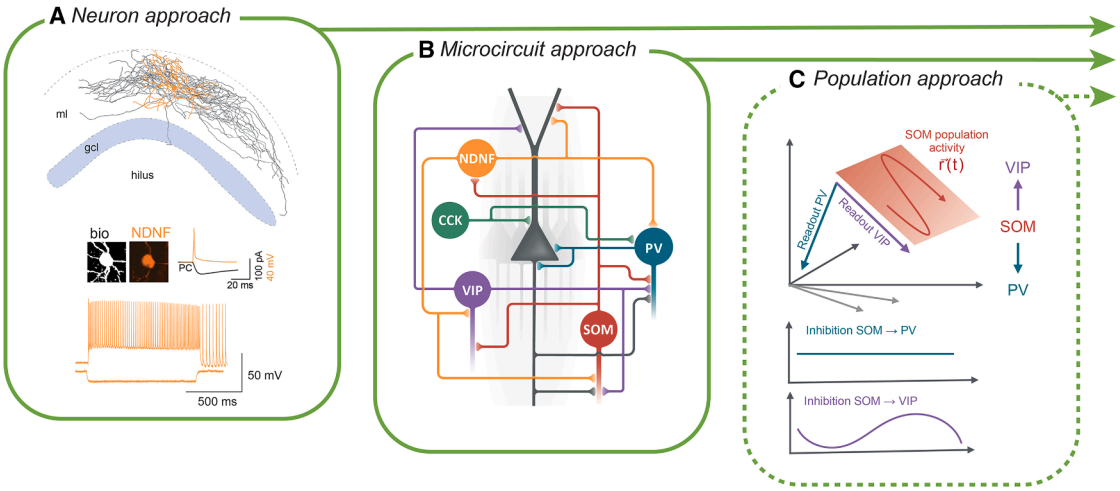

Figure 1. Major approaches toward interneuron function

(A) Schematic representation of the single-neuron approach based on the morphological, neurochemical, synaptic, and physiological properties of interneurons. Top: reconstruction of the somato-dendritic compartment (orange) and axon (black) of an intracellularly labeled neuron-derived neurotrophic factor-expressing interneuron (NDNFI) in the molecular layer (ml) of the dentate gyrus. Bottom: antibody labeling against NDNF (orange) of the same cell as shown above. Long-lasting negative and positive current injections induce a hyperpolarization and generation of action potentials, respectively (note persistent discharges).

(B) Circles and triangles represent the soma of interneurons and pyramidal cells, respectively. Thick lines depict dendrites, and thin lines depict axonal projections. SOM-, PV-, CCK-, VIP-, and NDNF-expressing interneuron classes are shown.

(C) Communication subspace analyses offer a fine-grain view of circuit interactions. Interneuron populations (e.g., SOMIs) may communicate different features of population activity to downstream targets (e.g., VIP- and PV-expressing interneurons, pyramidal cells) by means of target-specific synaptic projection patterns (readout vectors).

SOM, somatostatin; PV, parvalbumin; CCK, cholecystokinin; VIP, vasoactive intestinal peptide; gcl, granule cell layer.

relevance for our understanding on how cellular compartment-specific inhibition and connectivity motifs of the various interneuron types contribute to brain functions. On that basis, the new IN-CODE consortium funded by the German Research Foundation (DFG) argues that a third major approach with complementary strengths and limitations is required to illuminate functional aspects beyond the purview of the single-neuron and microcircuit fields. This notion is particularly motivated by some key gaps in our current knowledge. First, interneuron research is arguably *the* testbed for determining how much we can learn about brain function by an approach centered on neuronal identity. The core concept here is that circuits are composed of distinct elements with dedicated roles. However, the appropriate level of detail and the relevant parameters used to define interneuron types are far from self-evident. Moreover, we face a distinct (and growing) disconnect between research employing multi-omic approaches, which routinely define dozens of interneuron types,[1] and the typical *in vivo* experiments addressing the cardinal interneuron types identified by single neurochemical marker gene expression. We therefore still do not have a well-informed understanding of which granularity of interneuron categorization is best suited to explain a given brain function, let alone different brain functions combined.[2] Second, a related set of open questions pertains to the role of within-type heterogeneity. In the simplest interpretation, which is particularly prevalent in the microcircuit approach, the operational assumption is that all interneurons of a given type have similar properties and perform the same functions (Figure 1B). This is in stark contrast to the fact that cellular and synaptic properties, as well as *in vivo* activity patterns, of interneuron types are invariably heterogeneous. In fact, the variability of both sensory responses and their experience-dependent changes is often similarly pronounced within a given interneuron type and across different types. In part, this may relate to insufficient granularity of interneuron type definition, but variability is arguably an inherent feature of any neuronal encoding.[3] Finally, a further conspicuous gap of the conventional microcircuit approach is that recordings from different interneuron types are typically performed sequentially rather than in parallel. Apart from a few notable exceptions (e.g., Bugeon et al.[2] and Geiller et al.[4]), we currently know little about the combined roles of interneuron types (cooperation) in the processing and encoding of information in local circuits.

How do we deal with these challenges? Researchers that address information encoding in pyramidal cells have overcome similar issues by changing their focus from the neuron and microcircuit approaches to the larger-scale population level. This "population approach" emphasizes the collective nature of information processing and neural encoding in which the activity of individual cells provides a limited window into neuronal computations, because the relevant task variables are encoded in correlated variability among cells and are therefore difficult to reveal on the single-cell level. Instead, the population approach proposes that neuronal computation should be understood in terms of a set of underlying collective variables and their dynamics rather than the individual cell. Here, we aim to illustrate the usefulness of adding the population approach to the arsenal of interneuron research-related methodologies

and to provide entry points for achieving a better understanding on interneuron cooperativity for the encoding of behavioral variables in neuronal networks.

A population approach to interneuron function

Although for a long time the relative sparsity of interneurons in cortical circuits precluded the obtainment of the large datasets required for population-level analyses, recent methodological developments have made large-scale recordings of even less-abundant interneuron types in behaving animals feasible (e.g., Bugeon et al.[2] and Geiller et al.[4]; Figure 1C). This raises important questions: which approaches in population-level analysis are best suited for interneuron research, and how can we integrate the many existing and emerging insights we have gained from the neuron and microcircuit approaches into population-level analysis?

How do interneurons encode information?

Our view of the precision with which interneurons operate within cortical circuits to control behavioral output has evolved over the past decades. In contrast to previous views that interneurons have relatively unspecific circuit effects by providing a "blanket of inhibition" to large populations of pyramidal cells, several lines of experimental evidence indicate that interneuron types operate at higher precision. For instance, they show highly structured connectivity. PV cells in the visual cortex express higher mutual connection probability to pyramidal cells with similar sensory selectivity,[5] and inhibitory outputs of PV cells in the subiculum are non-randomly distributed, forming reciprocal microcircuit motifs that promote head-direction tuning.[6] Such highly structured connectivity indicates that interneurons support specific computations across brain areas.

However, it is often challenging to assess the information carried by interneurons by means of classical measures of tuning, because they frequently display mixed selectivity. Interneurons are, thus, typically tuned to several external and internal variables, e.g., reward locations, punishment, sensory stimuli, and behavioral states. Such mixed selectivity is more appropriately analyzed using a population approach, which interprets the collective activity as a high-dimensional population vector whose entries are the activities of individual cells. In simple behavioral tasks, the average population activity vector often lies on a low-dimensional surface, or "manifold" that occupies only a small fraction of the high-dimensional space of all possible neuronal activity patterns. Examining the "manifold geometry" of different interneuron types, as well as their changes in relation to behavioral states or in relation to experience, can reveal how interneuron populations encode the task space.

If the manifold is not strongly curved, changes in sensory stimuli, expected reward locations, or task dimensions (e.g., rule switching) cause changes along certain encoding axes. Projecting the population vector onto decoding axes then allows us to decode tasks, the context in which the animal is situated, and behavioral variables. Thus, "decoding" provides a simple way to quantify the information content on the population level. Moreover, comparative decoding of behavioral variables based on the activity of the different interneuron types and pyramidal cells will provide information on their contribution to the encoding of external features. Likewise, intermittent silencing of interneuron populations will be informative on how a given interneuron type affects the geometry of pyramidal cell manifolds and thereby will enlighten their individual role in the encoding of behavioral variables during task performance. Functionally, mixed selectivity could be particularly powerful for interneurons, because their lower cell numbers may well require a denser code than the sparse encodings often observed in pyramidal cells. Moreover, given that many interneurons express receptors for neuromodulators, the tuning of individual interneurons might be determined by nonlinear interactions of task variables such that the input related to one task variable can qualitatively change the cellular tuning to another variable.[7] Such a mechanism may be important for flexible reorganization of the neuronal circuit and the behavior in response to task changes. Nonlinearities in interneuron responses can be uncovered and quantified by visualizing the geometry of the population activity as a function of different task variables. Finally, while research over the last 20 years has highlighted the sensitivity of interneurons to a multitude of stimuli and task variables,[4] future research is needed to elucidate (1) whether this mixed encoding occurs at the level of individual interneurons and (2) whether it is indeed stronger than for pyramidal cells.

How do different interneuron types communicate?

Our current understanding of interactions between different neuron types is predominantly based on the microcircuit approach, in which neurons of a given type are treated as a homogeneous operational group with uniform impact on their diverse targets. On this coarse level, the impact of a given cell type on another can be assessed by quantifying the effect of its optogenetic manipulation. However, on a more fine-grained level, it remains unclear whether interneurons send the same information to all their targets. The propagation of information between two neuronal populations depends on the geometric alignment of the activity vector in the sending population with the structure of the functional projection (connectivity vector) between the two populations ("communication subspaces"). For instance, activity along axes that are orthogonal to this communication subspace does not influence the receiver and can therefore remain private.[8] Importantly, different downstream populations can read out different axes (dimensions) of neural activity, allowing the sending population to multiplex several information streams at once and transmit them differentially to different downstream target cell populations that are responsive to specific dimensions. Communication subspaces between different neural populations can be obtained by identifying dimensions in the activity of the sending population that best explain the activity in the receiving population, or when the two populations are mutually coupled, by fitting recurrent neural networks to their activity. The application of these methods to the communication between different interneuron types or between interneurons and pyramidal cells could expand our view of the synergistic interactions of all these cell types within the microcircuit. Furthermore, these results should be interpreted in light of the pronounced heterogeneity in interneuron

complements across cortical areas and even across species.

Technical challenges

Population-level analyses were originally applied to data from large-scale pyramidal cell populations. Their application to interneurons is more challenging. Interneurons form, in total, only 15%–20% of the entire cortical neuron population. Accordingly, cell numbers are often low and variable across interneuron types. Moreover, different interneuron types also discharge at different times and rates. As, for instance, decoding accuracy strongly depends on both neuron numbers and firing rates, it remains unclear whether the information content of the various interneuron types should be compared using matched cell numbers or whether it should be normalized to their tissue prevalence or the number of action potentials they emit.

Deeper insights into communication among the different types of interneurons will require the simultaneous recording of several identified interneuron types during task performance. Luckily, recent methodological breakthroughs have paved the way to address this goal, including the combination of calcium imaging of population activity with post hoc neurochemical or genetic neuron identification.[2,4] Moreover, the combinatory expression of interneuron type-specific enhancer adeno-associated viruses encoding for calcium indicators and activity modulators[9] allows for the examination of mutual communication among interneuron types and their cooperative roles in information processing and encoding.

Despite these technical challenges, comparative population-level analyses across different interneuron types as outlined above are an important first step toward linking the population approach to the existing body of work at the single-neuron and microcircuit levels. By correlating decoding accuracy, population geometry, communication subspaces, or dimensionality with neuronal identity, we can obtain a refined and expanded view of the function of individual interneuron types and their synergistic interactions within the circuit.

Bridging population codes to molecular and cellular signatures

How can we connect single-cell molecular and physiological signatures of interneuron types and population-level analyses? Despite the combinatorial coding of individual interneurons, recent studies support a "division of labor" among interneuron types during task performance, such as a strong correlation with place fields for PV cells, with specific contexts for SOM and rewards for hippocampal VIP cells.[10] Thus, we can assume that the accuracy of decoding latent variables correlates with the expression of a molecular marker. Linear decoders not only provide accuracy but also assign each neuron a decoding weight that characterizes how much this neuron contributes to the decoder. Given that VIP interneuron activity correlates with reward, a decoder of reward, trained on the recorded activity of identified interneurons in a pan-inhibitory mouse line, should consistently assign larger weights to VIP interneurons. Such an approach could be applied broadly. In fact, all linear population-level methods assign weights to individual neurons, which can be correlated with a broad variety of neuronal properties. By correlating single-cell characteristics (e.g., physiological properties or the transcriptome) with decoding weights, we could identify functional markers or signatures (e.g., spatial selectivity, generalization) that are specific and sensitive. These might then allow us to effectively observe and interfere with selected populations of neurons that encode a particular type of information, such as with holographic manipulation of the activity of selected neurons in behaving animals. Similarly, correlating connectivity with the factor loadings of dimensionality-reduction methods such as principal-component analysis may allow us to test whether connectivity motifs contribute more strongly to population dynamics. However, these prospective applications will need considerable future developments. One future challenge will be to build joint statistical models for linking *in vivo* population-level recordings and *ex vivo* identified molecular and cellular characteristics to thereby bridge between population codes and single-cell characteristics.

The utility of these analyses is not limited to the adult rodent brain, which is mostly discussed here. Indeed, these analyses should also be carried out longitudinally along the developmental trajectory in order to determine how population-level encoding by interneurons evolves during ontogeny. Furthermore, performing analogous analyses in models of neuropsychiatric disorders has a good potential for providing novel, population-level entry points into their pathophysiology. Finally, compared to mice, the human cortex has a 2.5-fold larger interneuron population and 10-fold higher mutual interneuron connectivity. In fact, this finding strongly underscores the importance of these neurons for cortical computations and cognition in an interspecies perspective. Thus, a major future challenge will be to obtain *in vivo* population-level recordings and single-interneuron characteristics in human cortex to test whether statistical models can explain and predict functional cooperativity of interneuron types across species. The proposed future work will support our understanding of not only the neuronal mechanisms underlying human cognition but will also identify links between inhibitory dysfunctions and neuropsychiatric disabilities, opening potential avenues for novel therapies.

ACKNOWLEDGMENTS

We would like to thank Dr. Aurore Cazala for the construction of the figures and Dr. Mark Eyre for comments on earlier versions.

AUTHOR CONTRIBUTIONS

M.B., J.R.G., J.L., and H.S. contributed to the conceptualization of this NeuroView. All authors contributed to the writing and proofreading of the paper with major roles of M.B., J.R.G., J.L., H.S., C.L., I.V., and C.M. in reviewing and editing.

DECLARATION OF INTERESTS

All authors declare no competing interest.

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